Replacing 3D Convolutions with Factorized Spatial Temporal Attention for Soccer Ball Action Spotting

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***Abstract*—Ball action spotting in soccer videos aims to automatically identify and temporally localize events such as passes and ball carries. This task requires modeling both visual information and temporal relationships between video frames. This paper presents a Transformer based approach for the SoccerNet Ball Action Spotting Challenge 2023, investigating the use of factorized spatial temporal attention as an alternative to the 3D convolutional temporal encoder of an existing solution. The proposed architecture uses an ImageNet pretrained EfficientNetV2-B0 backbone to extract spatial features from groups of sampled video frames. These features are projected into a shared embedding space and organized as spatial tokens. A temporal Transformer models the evolution of features independently at each spatial location, followed by a spatial Transformer that captures relationships between locations at each time step. The resulting representations are aggregated to predict the two target event classes: passes and drives. The model is evaluated using the seven fold cross-validation protocol adopted by the original solution. It achieves a mean mAP@1 of 77.45%, with a standard deviation of 5.47 percentage points across folds. This result is 1.61 percentage points below the 79.06% cross validation mAP@1 reported for the original CNN based approach. The findings provide an initial assessment of factorized spatial temporal attention for ball action spotting and highlight its potential as an alternative to convolutional temporal modeling.**

***Index Terms*—Ball action spotting, soccer video analysis, temporal action localization, 3D convolutional neural networks, Transformers, video understanding.**

I. Introduction

The improvements in deep learning and increase computing power permitted considerable progress in the field of sports video understanding. In particular sport video understanding field has become more important area for computer vision researcher [1] [2] [3] moving from easier task like object detection to more demanding challenges like temporal action spotting.

Temporal action spotting is a computer vision task that identifies the exact timestamp when a specific action or event begins or occurs in an untrimmed video, without needing to outline its full start and end duration.

To advance research in soccer video understanding the SoccerNet Challenge was proposed at CVPR 2023. The challenge focused on 7 tasks divided into three categories: Broadcast video understanding, Field Understanding, and Player understanding. We will focus on the first edition of Ball action spotting challenge included in the first category.

A. Cioppa et al. [4] define ball action spotting challenge as localizing the exact time when two types of actions related to the soccer ball occur: pass and drive. This task pose a unique challenge due to the dynamic and complex nature of the game. During a match players move rapidly, often obscuring each other, making it difficult to accurately track their movements. In this challenge a precise timestamp is set for this actions:

- **pass event**: the timestamp correspond to the moment when the ball leaves a player's foot
- **drive event**: the timestamp corrrespond to the moment when a player control the ball

The dataset consist of 7 games with 11,041 annotated timestamps with 2 games not publicy available for the challenge set. The recordings of the games are untrimmed and provided by Footovision. Other details of the challenge can be found at SoccerNet Ball Action Spotting challenge.

The main contributions of this work are:

- We investigate a Transformer based temporal encoder for ball action spotting, replacing the 3D convolutional component of an existing architecture.
- We describe an architecture that combines a shared 2D visual feature extractor with self-attention based temporal feature aggregation.

II. Related Work

A. Soccer Action Spotting

Soccer action spotting has received increasing attention in computer vision, particularly with the introduction of the SoccerNet benchmark [4]. The benchmark provides broadcast soccer videos annotated with temporally localized events, enabling the development and evaluation of methods for detecting relevant moments in long, untrimmed recordings.

The SoccerNet 2023 challenges further expanded this research direction by introducing several tasks addressing broadcast video, field, and player understanding [4]. Within this framework, Ball Action Spotting focuses specifically on identifying two ball related events: passes and drives. Each event is represented by a timestamp corresponding to a precisely defined interaction with the ball. This formulation differs from broader

action spotting tasks because the target events are closely related to changes in ball possession and control.

The temporal precision required by this task makes the extraction of information from neighboring frames particularly important. A single frame may not provide sufficient context to distinguish the relevant interaction, motivating architectures that combine spatial feature extraction with temporal modeling.

B. Ball Action Spotting

The SoccerNet Ball Action Spotting Challenge 2023 attracted several approaches based on deep learning architectures and different training strategies. Among them, Baikulov achieved first place with an efficient architecture combining 2D and 3D convolutional operations [5].

The proposed solution adopts a slow fusion strategy. A shared 2D convolutional backbone independently extracts visual features from groups of neighboring grayscale frames. These feature representations are subsequently processed by a 3D convolutional encoder, which captures spatiotemporal information. Generalized mean pooling is then applied to aggregate the resulting features before the final classification stage.

An important aspect of this approach is the separation of spatial and temporal feature extraction. By applying 2D convolutions before 3D convolutions, the architecture first extracts visual information from individual frame groups and subsequently integrates information across time. This design provides an effective balance between spatial representation learning and temporal feature extraction.

Beyond the model architecture, Baikulov's solution employs a multistage training strategy. This includes pretraining on the SoccerNet Action Spotting dataset, transfer learning, and finetuning with longer temporal sequences. The solution also introduces an out-of-fold predictionbased sampling strategy to increase the probability of selecting challenging training examples. These choices highlight the importance of the training procedure, in addition to the architecture itself, for achieving strong spotting performance.

Another notable approach was proposed by Wang et al. [6], who obtained second place in the 2023 challenge with Boosted Model Ensembling (BME). Their method builds upon the E2E-Spot architecture [7] and focuses on generating and combining multiple candidate models. The candidate models differ in their feature extraction architectures and training configurations. An iterative selection procedure is then used to identify suitable ensemble members and assign their respective weights according to their validation performance.

These two solutions illustrate different strategies for improving ball action spotting: the use of an efficient spatiotemporal architecture and the combination of diverse models through ensemble learning. In particular, Baikulov's slow fusion architecture provides the reference for the approach investigated in this work. We explore an alternative temporal modeling strategy by replacing its 3D convolutional encoder with a Transformer based encoder while retaining the shared 2D visual feature extractor.

C. Convolutional Temporal Modeling

Convolutional neural networks are widely used in video understanding because they can learn hierarchical visual representations directly from image data. A conventional 2D CNN applies convolutional filters across the spatial dimensions of an image, extracting features such as edges, textures, and higher level semantic patterns.

However, video data also contains temporal information that cannot be fully represented by processing frames independently. Three-dimensional convolutional neural networks address this limitation by extending convolutional operations to the temporal dimension. Their filters operate across both spatial and temporal dimensions, allowing the network to learn local motion patterns and interactions between neighboring frames.

An alternative to applying 3D convolutions throughout the entire network is the slow fusion strategy. In this approach, spatial features are initially extracted using 2D convolutions and are progressively integrated through subsequent temporal processing stages. This reduces the need to perform full spatiotemporal convolutions at the earliest network layers while retaining the ability to learn temporal representations.

The slow fusion architecture employed by Baikulov [5] follows this principle by combining a shared 2D feature extractor with a 3D convolutional temporal encoder. The resulting architecture serves as the convolutional reference for our investigation.

D. Transformer Based Temporal Modeling

Transformers [8] were originally introduced as an attention based architecture for sequence modeling. Their central component is the self-attention mechanism which enables each element of a sequence to incorporate information from other elements through interactions between query, key, and value representations.

Unlike convolutional operations, which aggregate information through locally defined receptive fields, self-attention computes relationships between elements across the input sequence. This allows the model to capture dependencies between temporally separated representations without requiring information to pass exclusively through successive local operations.

Since Transformers do not inherently encode the ordering of input elements, positional information is incorporated into the sequence representations. Positional embeddings provide the model with information about the relative or absolute positions of individual elements, allowing temporal order to influence the attention computation.

In video understanding, this mechanism can be applied to sequences of visual feature representations extracted from individual frames or groups of frames. A temporal encoder can then learn relationships between these representations and aggregate the resulting information for video level classification.

In this work, we investigate this approach specifically for ball

action spotting. In contrast to approaches that globally pool spatial feature maps before temporal modeling, the proposed architecture retains a coarse spatial grid. Temporal attention is first applied independently at each spatial location, followed by spatial attention at each temporal position. This factorized design allows the model to capture both temporal evolution and spatial relationships. This provides an alternative to the convolutional temporal modeling strategy employed by the reference architecture.

III. Proposed Method

A. Architecture Overview

The proposed model is a single stage architecture designed to detect ball related actions by jointly exploiting spatial appearance and temporal information. It follows a two part feature extraction strategy: a convolutional backbone extracts visual representations from short groups of consecutive frames, while a factorized Transformer models temporal dependencies and spatial relationships within the resulting feature maps.
Given an input video segment containing $T = 15$ grayscale frames, the frames are divided into five non overlapping groups of three consecutive frames. Each group is processed independently by a shared 2D convolutional backbone. The resulting feature maps are projected to a common channel dimension and arranged as a spatio temporal feature tensor.
Unlike approaches that aggregate spatial information before temporal modelling, the proposed architecture preserves a spatial representation throughout the Transformer stages. Temporal attention is first applied independently at each spatial location. Then, spatial attention enables information exchange between different locations within each time step. This factorized design allows the network to model both the temporal evolution of visual features and their spatial relationships.
The final representation is obtained by averaging the encoded features over the temporal and spatial dimensions. A linear classification layer then produces the predictions for the two ball action categories considered in the 2023 challenge: pass and drive.

B. Frame Grouping and 2D Feature Extraction

Let the input video segment be represented by

$$X \in \mathbb{R}^{B\times T\times H\times W},$$

where B is the batch size, $T = 15$ is the number of input frames, and H and W denote the frame height and width.
To exploit short term visual information while maintaining the computational efficiency of a 2D backbone, consecutive frames are grouped into stacks of three. The input is therefore reshaped into five groups:

$$X_s \in \mathbb{R}^{(B\cdot 5)\times 3\times H\times W}.$$

Each group is treated as a three channel input and processed by the same 2D convolutional backbone. The backbone extracts a feature map for each group, producing a representation with 192 channels and a spatial resolution of 23×40 in the configuration shown in the implementation.
A 1×1 convolution followed by batch normalization and a SiLU activation is used to project the extracted features to a 192-dimensional representation. The projected feature maps are then reshaped to recover the temporal dimension:

$$F \in \mathbb{R}^{B\times 5\times 192\times 23\times 40}.$$

At this stage, each of the five temporal positions is represented by a spatial feature map. The shared convolutional backbone provides visual descriptors, while the subsequent Transformer stages explicitly model their spatio temporal relationships.

C. Spatial Feature Reduction

Directly applying attention to the full spatial resolution would result in a large number of tokens and substantial computational cost. To make attention based processing more efficient while retaining spatial information, adaptive average pooling is applied independently to each feature map.
The spatial dimensions are reduced from 23×40 to 6×10:

$$F_p \in \mathbb{R}^{B\times 5\times 192\times 6\times 10}.$$

The resulting grid contains 60 spatial locations per time step. Each location is represented by a 192-dimensional feature vector. Importantly, this operation does not collapse the entire feature map into a single vector; it preserves a coarse spatial layout that is subsequently used by the attention modules.

D. Factorized Spatial Temporal Transformer

The central component of the proposed architecture is a factorized Transformer encoder composed of two sequential attention stages: temporal attention and spatial attention.

Temporal attention. First, the feature tensor is reorganized so that each spatial location is associated with a sequence of five feature vectors, one for each temporal group. The tensor is reshaped as

$$F_t \in \mathbb{R}^{(B\cdot 60)\times 5\times 192}.$$

Consequently, each spatial location is processed as an independent temporal sequence. This design allows the model to learn how the visual representation at a particular location changes across the input segment, without initially mixing information from different spatial locations.
A learnable temporal positional embedding is added to each sequence to encode the ordering of the five temporal positions. The resulting representation is processed by a Transformer encoder consisting of two layers. Each layer uses multihead self attention with eight attention heads, a feed forward dimension of 768, and a dropout probability of 0.1.
The temporal encoder therefore models dependencies between the five frame groups independently at each of the 60 spatial locations.

Spatial attention. After temporal encoding, the tensor is rearranged to restore the temporal and spatial dimensions:

$$F'_t \in \mathbb{R}^{B\times 5\times 60\times 192}.$$

Spatial attention is then applied independently to each temporal position. For each of the five time steps, the 60 spatial feature vectors form a sequence:

$$F_s \in \mathbb{R}^{(B\cdot 5)\times 60\times 192}.$$

A learnable spatial positional embedding is added to represent the locations within the 6×10 feature grid. The resulting sequence is processed by a Transformer encoder with one layer, eight attention heads, a feed forward dimension of 768, and a dropout probability of 0.1.
This second stage enables the model to exchange information between different spatial locations. In combination with the preceding temporal encoder, it allows the network to represent both how visual features evolve over time and how different regions of the scene relate to one another.

E. Feature Aggregation and Classification

Following the spatial Transformer, the encoded representation is reshaped to

$$F_{\text{out}} \in \mathbb{R}^{B\times 5\times 60\times 192}.$$

The temporal and spatial dimensions are averaged to obtain one feature vector per input segment:

$$z_b = \frac{1}{5\cdot 60}\sum_{t=1}^{5}\sum_{s=1}^{60} F_{\text{out}}[b,t,s,:].$$

The resulting representation has 192 dimensions and summarizes the information extracted across all temporal groups and spatial locations. Finally, a fully connected layer maps this representation to the output classes:

$$\hat{y}_b = W z_b + b,$$

where the output dimension corresponds to the two ball action categories, pass and drive. The resulting scores are used to predict the presence of the corresponding actions in the input segment.

IV. Experimental Setup

A. Dataset and Task

The experiments are conducted in the context of the SoccerNet Ball Action Spotting Challenge 2023 [4]. The task consists of identifying two types of ball related actions: passes and drives. A pass corresponds to the moment when the ball leaves a player's foot, whereas a drive corresponds to the instance when a player controls the ball.
The dataset consist of broadcast football videos annotated with temporal action labels. The objective is to predict the temporal locations of the annotated actions from the video frames.

B. Input Representation

The proposed model receives a sequence of 15 grayscale frames as input. Consecutive frames are grouped into stacks of three, resulting in five temporal groups. Each group is processed by the shared 2D convolutional backbone, and the resulting feature maps are passed to the factorized spatial temporal Transformer.

[TO BE COMPLETED: Specify the frame sampling rate, input resolution, temporal distance between sampled frames, and any image preprocessing or normalization applied.]

C. Training Configuration

The proposed model was trained following the training procedure of the original Ball Action Spotting Challenge solution by Baikulov [5]. The original training configuration was retained, including the optimizer, learning rate schedule, number of epochs, and loss function.
The only modification concerned the batch size, which was reduced from 2 to 1 due to GPU memory constraints. While the original implementation was trained on an NVIDIA GeForce RTX 3090 Ti, our experiments were conducted on an NVIDIA GeForce RTX 3080. This adjustment allowed the model to be trained within the available hardware resources while preserving the remaining training settings.
The experimental evaluation considers the model obtained directly after the training procedure. No additional finetuning stages, training phases, or post processing operations were applied to the predictions.
The model was trained using the configuration adapted from Baikulov's reference implementation [5]. The input videos were represented by sequences of 15 grayscale frames, sampled with a temporal step of two frames and grouped into five stacks of three consecutive frames. Each frame was resized and padded to a resolution of 1280 × 736.
The model was optimized using AdamW, with a base learning rate of 3×10^{-4} and a batch size of 1. Training employed mixed-precision computation and an epoch size of 6,000 samples. The training schedule consisted of two stages: a six epoch warmup stage followed by a 30 epoch training stage. Focal loss was used, with a focusing parameter $\gamma = 1.2$.
An exponential moving average of the model parameters was maintained with a decay factor of 0.999. The EfficientNetV2-B0 backbone was initialized with pretrained ImageNet-1K weights and was not frozen during training. The batch size was reduced from the reference configuration to accommodate the available GPU memory.

D. Dataset Partitioning and Training Protocol

We adopted the seven fold cross-validation strategy of Baikulov [5]. Each labelled game was assigned to one fold. For each fold, six folds were used for training, while the remaining fold was reserved for validation. This procedure was repeated seven times, resulting in seven fold specific models.
The proposed Transformer architecture was trained using this protocol. The original training configuration was retained, with the exception of the batch size, which was reduced from 2 to

1 to accommodate GPU memory constraints. No additional finetuning or postprocessing was performed.

E. Evaluation Metric

Performance is evaluated using mean Average Precision (mAP), computed with the official SoccerNet evaluation implementation for the Ball Action Spotting Challenge 2023. Following the evaluation procedure adopted by Baikulov [5], predictions for the publicly available test games are obtained using the corresponding out-of-fold models. The official challenge test set is not publicly available for independent evaluation; therefore, the results reported in this work correspond to the publicly available test data and should not be interpreted as official challenge results.

F. Transformer Based Temporal Modeling

To investigate the use of self attention for temporal feature modeling, we replace the original 3D convolutional encoder with a Transformer encoder while retaining the shared 2D visual feature extractor. The input video is divided into groups of three consecutive grayscale frames, and each group is processed independently by the 2D encoder. The resulting feature maps are projected to a common feature dimension of 192 channels.

Unlike the original architecture, which performs temporal modeling on spatial feature maps, the proposed approach first applies adaptive average pooling to each feature map, producing a sequence of five 192-dimensional feature vectors for a 15 frame input. Learned positional embeddings are added to the sequence to provide information about the temporal ordering of the feature vectors.

The resulting sequence is processed by a Transformer encoder consisting of two layers with eight attention heads and an embedding dimension of 192. Selfattention allows each temporal token to incorporate information from the other tokens in the input sequence. The encoded representations are then aggregated by temporal average pooling to obtain a single 192-dimensional video representation.

Finally, a linear classification layer maps this representation to the output action classes. This architecture replaces the original 3D convolutional temporal encoder and its subsequent feature projection and concatenation with a Transformer based temporal encoder and average pooling operation. The effect of this architectural change is evaluated using the same action spotting task and evaluation metric as the baseline.

G. Ball Action Spotting

The SoccerNet Ball Action Spotting Challenge was introduced to address the problem of temporally localizing ball related actions in broadcast soccer videos. Unlike general action spotting, ball related events such as passes and drives are often densely distributed and require precise temporal localization.

Among the solutions proposed for the 2023 challenge, Baikulov achieved first place with an efficient deep learning architecture combining two-dimensional and three-dimensional convolutional operations [5]. The proposed architecture follows a slow fusion strategy, in which a shared 2D convolutional encoder independently extracts visual features from groups of consecutive grayscale frames. These features are subsequently processed by a 3D convolutional encoder to capture temporal information. The resulting representations are aggregated through generalized mean pooling and used for multilabel action classification.

In addition to the model architecture, the solution introduced a multi stage training strategy involving pre training on the SoccerNet Action Spotting dataset, transfer learning, and finetuning with longer video sequences. The use of out-of-fold predictions for sample selection and an efficient video loading pipeline further contributed to the final performance. The solution achieved 86.47% mAP@1 on the challenge set [5].

Another notable approach was proposed by Wang et al., who obtained second place in the 2023 challenge with Boosted Model Ensembling (BME) [6]. Their method is based on the E2E-Spot baseline and focuses on improving performance through the generation and combination of multiple candidate models. These models differ in their training sample configurations, feature extraction architectures, and optimization strategies. The candidate architectures include RegNet and EfficientNet backbones incorporating Gate Shift Modules, while the training strategies consider AdamW and stochastic weight averaging.

Rather than relying on a conventional fixed ensemble, BME iteratively selects candidate models and determines their contribution according to improvements in validation mAP@1. The resulting ensemble achieved 86.37% mAP@1 on the challenge test set [6].

These approaches demonstrate the importance of both spatiotemporal feature extraction and training strategies for accurately spotting ball related events. In particular, the slow fusion architecture proposed by Baikulov provides the convolutional baseline considered in this work. We investigate an alternative approach to temporal feature modeling by replacing its 3D convolutional encoder with a Transformer encoder, while retaining the shared 2D visual feature extractor.

H. Evaluation Metrics

Performance is evaluated using mean Average Precision at a temporal tolerance of one second (mAP@1), following the official SoccerNet Ball Action Spotting Challenge 2023 evaluation protocol. Average Precision is computed independently for each event category, Pass and Drive, and mAP@1 is obtained by averaging the two class-specific AP values. The one-second tolerance reflects the high temporal precision required to localize ball-related events.

V. Results and Discussion

A. Cross-Validation Results

The proposed Transformer based architecture was evaluated using the seven fold cross-validation protocol. Each fold uses six games for training and one held out game for validation.

Model	CV mAP@1 (%)	Public test mAP@1 (%)
Baikulov basic training	79.06	84.26
Baikulov final model	80.49	87.04
Proposed Transformer	77.45	81.79

TABLE I
COMPARISON OF OUR PROPOSED TRANSFORMER WITH ORIGINAL CNN MADE BY BAIKULOV IN TERMS OF AVERAGE MAP IN THE SOCCERNET 2023 CHALLENGE.

The evaluation script computes mAP@1 independently for each held out game.

The model achieves a mean cross-validation mAP@1 of **77.45%**, with a standard deviation of **5.47 percentage points** across the seven folds. Individual fold results range from 69.97% to 83.25%, indicating some variation in performance across games.

For reference, Baikulov reports a cross-validation mAP@1 of **79.06%** for the original CNN based solution. The proposed Transformer therefore obtains a result 1.61 percentage points lower than the reported reference. This comparison provides an indication of the performance difference, although it should not be interpreted as a fully controlled architectural ablation.

B. Public Test Results

The public test evaluation follows the same out-of-fold prediction procedure as Baikulov's implementation. Predictions are generated using the two fold specific models that include the public test games, allowing the results to be compared under the same evaluation protocol.

The original CNN based solution reports **84.26%** mAP@1 on the public test set. The proposed Transformer achieves **81.79%**mAP@1.

Comparison with the Original Solution

Table I summarizes the reported cross-validation and public test results. The original CNN results are taken from Baikulov's repository. The Transformer results are obtained from the experiments described in this work.

The comparison shows how the proposed factorized spatial temporal Transformer performs relative to the original CNN based solution under the corresponding evaluation protocols. The results for Baikulov's two configurations are taken from the original repository. The basic-training and final-model results correspond to different training procedures. In particular, the final model includes additional long-sequence fine-tuning and inference procedures. Therefore, these results provide reference points rather than a controlled comparison of the temporal architectures.

VI. CONCLUSION

This paper investigated the use of factorized spatial temporal Transformer blocks for ball action spotting in soccer videos, focusing on the two event categories considered in the SoccerNet 2023 challenge: passes and drives. The proposed architecture retains a pretrained 2D EfficientNetV2-B0 backbone for visual feature extraction while replacing the original 3D convolutional temporal modeling with two sequential attention stages. The temporal Transformer captures changes in visual features over time at individual spatial locations, while the spatial Transformer models relationships between different locations within each temporal step.

The proposed model achieved a mean mAP@1 of 77.45% across seven cross-validation folds, with a standard deviation of 5.47 percentage points. Compared with the 79.06% cross-validation result reported by Baikulov for the original CNN based solution, this represents a difference of 1.61 percentage points. Although the proposed architecture does not exceed this reference result, it demonstrates that factorized spatial temporal attention can be applied to the task while retaining spatial information throughout temporal modeling.

Several aspects remain open for further investigation. In particular, a controlled comparison in which the original CNN and the proposed Transformer are trained under identical conditions would help isolate the effect of the temporal modeling architecture. Further experiments could also investigate the number of spatial tokens, the depth of the Transformer blocks, and alternative strategies for aggregating spatiotemporal features.

Overall, this work provides an initial exploration of Transformer based spatio temporal modeling for ball action spotting. The results offer a basis for further investigating attention based architectures for temporally precise event localization in soccer videos.

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